

Cohort, labels, preprocessing, architecture and optimiser are identical across all three arms

Cohort

740 patients
1,695 nodules
11,026 slices

Preprocessing

50% consensus ROI
256 × 256 px
identical in all arms

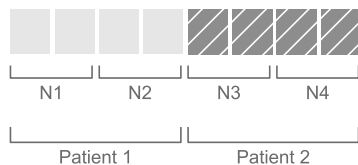
Backbone

ImageNet-pretrained
DenseNet-121
EfficientNet-B0

the partition unit is the only thing that changes

(a) Arm A — Patient-grouped

GroupKFold on patient



Fraction of test rows that share a

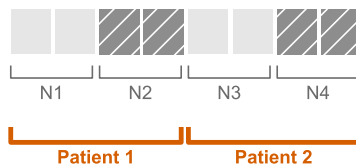
patient with the training fold 0.000

nodule with the training fold 0.000

neither unit crosses the boundary

(b) Arm C — Nodule-grouped

GroupKFold on nodule



Fraction of test rows that share a

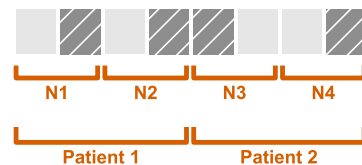
patient with the training fold 0.690

nodule with the training fold 0.000

patient route on, within-nodule route off

(c) Arm B — Random

KFold, no grouping



Fraction of test rows that share a

patient with the training fold 0.995

nodule with the training fold 0.973

both routes on

B - A total gap

light gray training slice

dark gray with diagonal lines test slice

C - A patient route

orange line unit split across training and test

B - C within-nodule route